1. **A. Basic Data Handling Commands:**

```
import pandas as pd

# Read CSV file
df = pd.read_csv("data.csv")

# Display data
print(df.head())      # Top 5 rows
print(df.shape)       # Dimension (rows, columns)
print(df.columns)     # Column names
print(df.tail())      # Last 5 rows
```

B. Hive Partition & Bucketing

```
 // 1. Create / use database
use shopping
// 2. Create collection (table equivalent)
db.createCollection("shopping1")
// 3. Insert sample data (since MongoDB doesn't load like Hive)
db.shopping1.insertMany([
  { id: 1, product: "Shirt", price: 500, category: "Clothing" },
  { id: 2, product: "Laptop", price: 50000, category: "Electronics" },
  { id: 3, product: "Shoes", price: 2000, category: "Footwear" }
])
// 4. View data
db.shopping1.find()
// 5. Partition equivalent (group by category)
db.shopping1.aggregate([
  { $group: { _id: "$category", items: { $push: "$$ROOT" } } }
])
// 6. Bucket equivalent (group into ranges based on id)
db.shopping1.aggregate([
```

```
  {
    $bucket: {
      groupBy: "$id",
      boundaries: [0, 2, 4, 10],
      default: "Other",
      output: {
        count: { $sum: 1 }
      }
    }
  }
])
```

2.A. Data Handling Operations

```
import pandas as pd
df = pd.read_csv("data.csv")
# Column names
print(df.columns)
# Rename column
df.rename(columns={"model year": "year"}, inplace=True)
# Bind rows
df2 = df.copy()
df_rows = pd.concat([df, df2]).reset_index(drop=True)
# Bind columns
df_cols = pd.concat([df, df2], axis=1)
# Missing values
print(df.isnull())
print(df.isnull().sum())
```

B. Feature Extraction (Correlation Heatmap)

```
import seaborn as sns
import matplotlib.pyplot as plt
corr = df.corr()
sns.heatmap(corr, annot=True)
plt.show()
```

3.A. Data Visualization (Iris Dataset)

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
df = pd.read_csv("iris.csv")
# Box plots
sns.boxplot(x='class', y='sepal length', data=df)
plt.show()
sns.boxplot(x='class', y='sepal width', data=df)
plt.show()
sns.boxplot(x='class', y='petal length', data=df)
plt.show()
# Histogram
sns.histplot(x='sepal length', data=df)
plt.show()
# Scatter plot
sns.scatterplot(x='petal length', y='sepal length', data=df)
plt.show()
```

B. LDA (Linear Discriminant Analysis)

```
 from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
from sklearn import datasets
data = datasets.load_iris()
X = data.data
y = data.target
lda = LinearDiscriminantAnalysis(n_components=2)
X_new = lda.fit(X, y).transform(X)
```

4.Feature Construction (Homeprices Dataset)

Dummy Encoding

```
import pandas as pd
df = pd.read_csv("homeprices.csv")
dummies = pd.get_dummies(df['town'])
dummies = dummies.drop(['west windsor'], axis=1)
df = pd.concat([df, dummies], axis=1)
df = df.drop(['town'], axis=1)
```

Label Encoding

```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df['town'] = le.fit_transform(df['town'])
```

Convert Numeric → Categorical

```
df['category'] = pd.cut(df['price'], bins=[0,100000,200000,300000])
```

5. Data Preprocessing (AutoMPG Dataset)

Remove Missing Values

```
import numpy as np

df.replace("?", np.nan, inplace=True)
```

Outlier Removal using IQR

```
Q1 = df['mpg'].quantile(0.25)

Q3 = df['mpg'].quantile(0.75)

IQR = Q3 - Q1

LL = Q1 - 1.5 * IQR

UL = Q3 + 1.5 * IQR

df = df[(df['mpg'] >= LL) & (df['mpg'] <= UL)]
```

Imputation

```
df.fillna(df.mean(), inplace=True)
```

Capping

```
df['mpg'] = np.where(df['mpg'] > UL, UL, df['mpg'])
```

6. Feature Extraction (PCA, LDA, SVD)

```
from sklearn.decomposition import PCA

from sklearn import datasets

data = datasets.load_iris()

X = data.data

pca = PCA(n_components=2)

X_pca = pca.fit_transform(X)
```

SVD

```
import numpy as np

U, S, Vt = np.linalg.svd(X, full_matrices=False)
```

7. HDFS Commands

```
# Start Hadoop
start-dfs.sh
start-yarn.sh
# Create directory
hdfs dfs -mkdir /hadooplab
hdfs dfs -mkdir /hadooplab/input
# Create file
echo "Hello Hadoop" > sample.txt
# Upload file
hdfs dfs -put sample.txt /hadooplab/input
# View file
hdfs dfs -cat /hadooplab/input/sample.txt
# Download file
hdfs dfs -copyToLocal /hadooplab/input/sample.txt
```

8. MapReduce Word Count

9. Hive Database Creation

```
use shopping
db.createCollection("shopping1")
db.shopping1.insertMany([
  { id: 1, product: "Shirt", price: 500 },
  { id: 2, product: "Laptop", price: 50000 },
  { id: 3, product: "Shoes", price: 2000 }
])
db.shopping1.find()
```

```
mongoimport --db shopping --collection shopping1 --type csv --file shop.txt --headerline
```

10. MongoDB Queries

```
// Create DB
use library
// Create collection
db.createCollection("books")
// Insert
db.books.insertOne({name:"DBMS", price:500})
db.books.insertMany([{name:"OS"}, {name:"CN"}])
// Find
db.books.find()
db.books.find({price:500})
// Update
db.books.updateOne({name:"OS"}, {$set:{price:300}})
db.books.updateMany({}, {$set:{available:true}})
// Delete
db.books.deleteOne({name:"CN"})
db.books.deleteMany({})
```